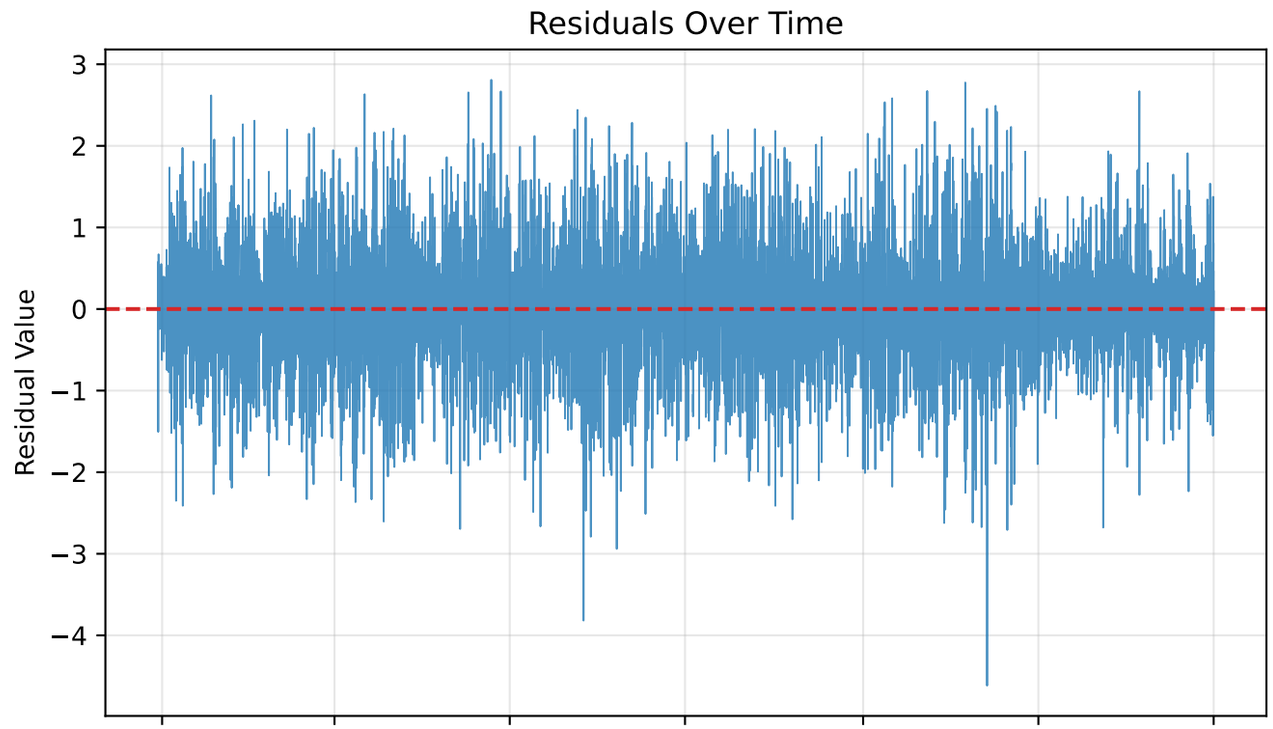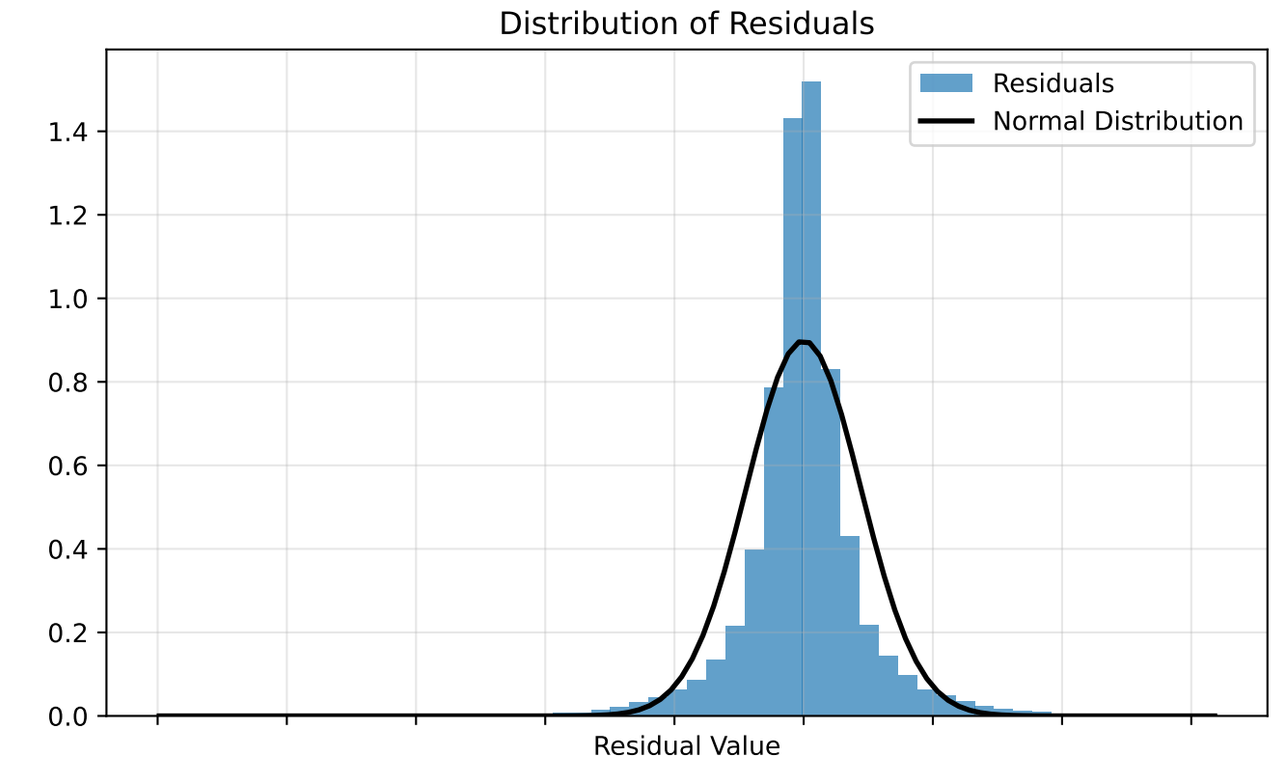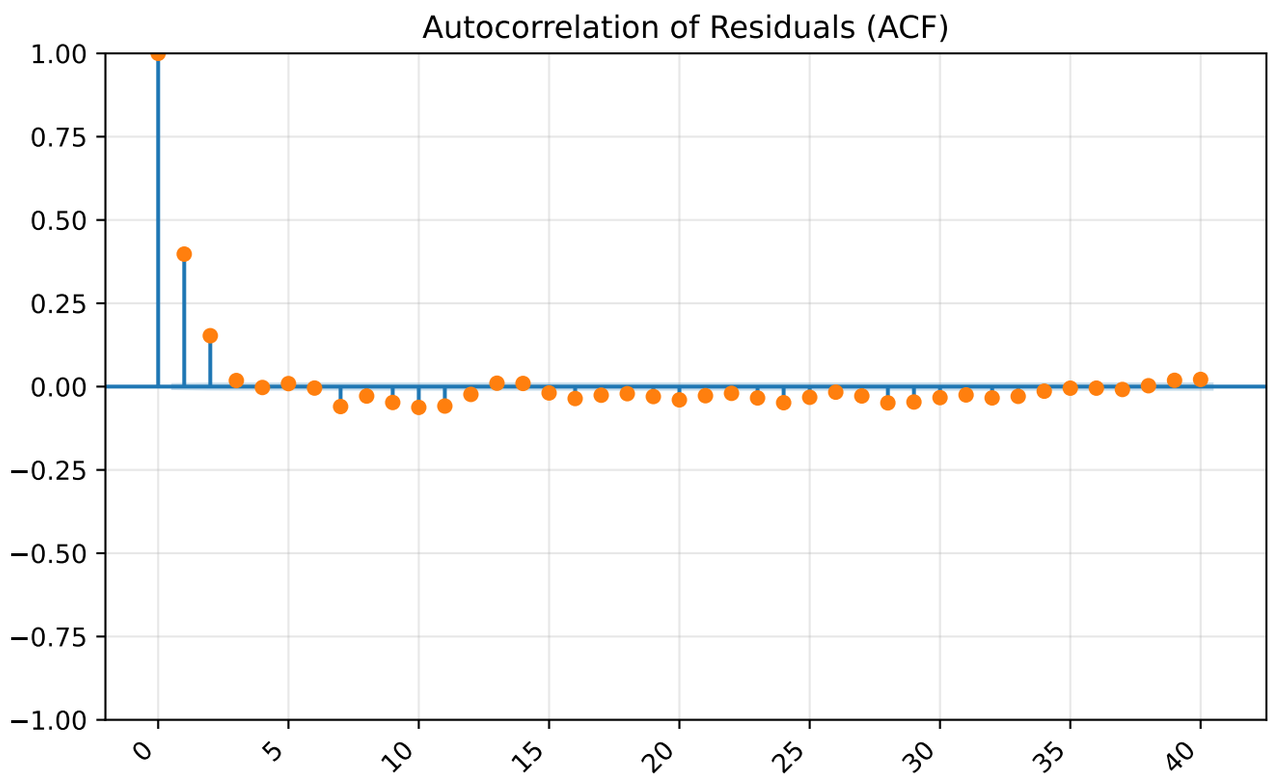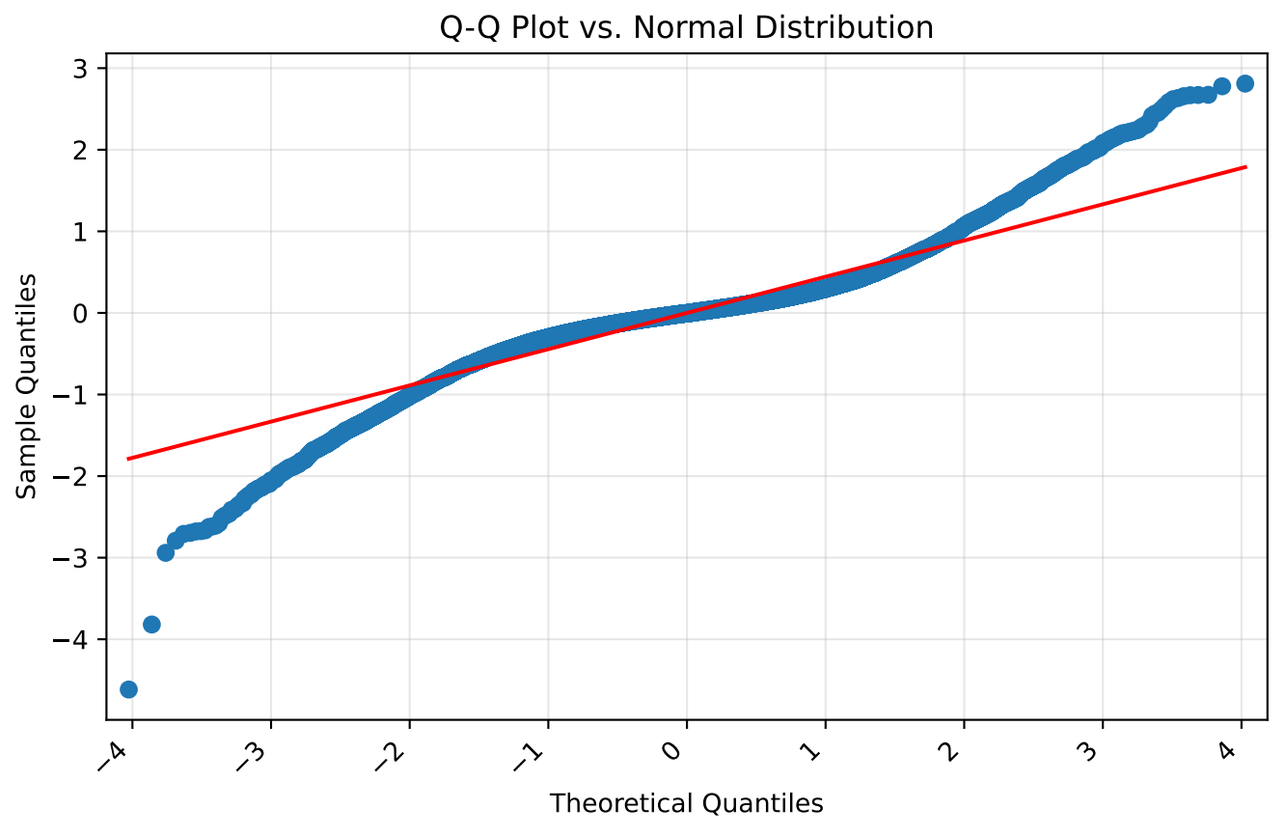Residual Diagnostics for 50Hertz (2024)